BIOCHEMISTRY -I (BIO202)

CURRENT FINAL PAPER SPRING 2023(25August)

SHORT QUESTIONS(3)

1) Types of lipoprotein

The lipoprotein particles include

1) chylomicrons (CM),
2) very-low-density lipoproteins (VLDL),
3) low-density lipoproteins (LDL), and
4) high-density lipoproteins (HDL).

2) Components of nucleotides

Nucleotides are composed of three main components:

1) a nitrogenous base (adenine, guanine, cytosine, thymine, or uracil),
2) a pentose sugar (ribose or deoxyribose), and
3) a phosphate group.

These components come together to form the building blocks of nucleic acids like DNA and RNA.

3) Write three modified bases?

These ***unusual bases***(also called ***modified bases***) include

1) Dihydrouracil
2) Pseudouridine
3) Thymine.

4) Functional group of pyrimidines and purines?

The most important functional groups of pyrimidines and purines are

1) ring nitrogens,
2) carbonyl groups, and
3) exocyclic amino groups.

5) Three names of polyunsaturated fatty acid

Polyunsaturated fatty acids contain more than one double bond • These include:

1) Linoleic acid (18: 2Δ9, 12)
2) α-Linolenic acid (18 : 3Δ9, 12, 15)
3) Arachidonic acid series (20 : 4Δ5, 8, 11, 14)

6) Enzyme activity in three steps

• Enzyme-catalyzed reactions have three basic steps:

1) binding of substrate: E+S ↔ ES
2) conversion of bound substrate to bound product: ES ↔ EP
3) release of product : EP ↔ E+P

7) Mention inorganic cofactors

The inorganic cofactor involved in coenzymes include:

a. Zinc (Zn)
b. Iron (Fe)
c. Magnesium (Mg)
d. Copper (Cu)
e. Manganese (Mn)

8) Name Of three pyrimidine

•Pyrimidines include:

A. Cytosine C—in both DNA and RNA
B. Thymine (T)—only in DNA

C. Uracil (U) —only in RNA
   a. Cytosine is 2-oxy-4- amino-pyrimidine
   b. Thymine is 2,4-dioxy-5- methyl-pyrimidine
   c. Uracil is 2,4-dioxypyrimidine

9) Pepsin PH

pepsin, a digestive enzyme in the stomach, is maximally active at pH 2.

10) Concentration of tRNA in cell

tRNAs compose roughly 20% of total cellular RNA • There are at least 20 species of tRNA molecules in every cell.

11) Number of carbon atom in Stearic acid

Stearic acid is a saturated fatty acid, meaning it does not contain any double bonds in its carbon chain. It consists of 18 carbon atoms in total.

12) Monosaccharides

Monosaccharides (simple sugars) are those which cannot be hydrolyzed further into simpler forms**.**
e.g: Glucose

13) Polysaccharides with two example

These are complex carbohydrates made up of multiple monosaccharide units. ***Examples*** include starch (energy storage in plants), glycogen (energy storage in animals), and cellulose (structural component in plant cell walls).

14) How fireflies produce light?

The light produced by fireflies is the result of a reaction involving the protein luciferin and ATP, catalyzed by the enzyme luciferase.

15) Result of reduction of glutamate

• Removal of glutamate residue from LTC4(4 is subscript) forms LTD4 • LTD4 on removal of glycine becomes LTE4

16) Hypercholesterolemia

"hypercholesterolemia," which is a condition characterized by high levels of cholesterol in the blood. This condition can increase the risk of heart disease and other cardiovascular issues.

17) Effect of increase in GC base pairs

• The higher the content of GC base pairs, the higher the melting point of the DNA. • This is because GC base pairs, with three hydrogen bonds, require more heat energy to dissociate than AT base pairs.

18) Enantiomers and diastereomers

Stereoisomers that are mirror images of each other are called ***enantiomers***

Pairs of stereoisomers that are not mirror images of each other are called ***diastereomers***

19) Diff. B/w Tyrosine Tryptophan and phenyl alanine on ring structure

Tyrosine: Amino acid with phenyl group and hydroxyl (-OH).

Tryptophan: Amino acid with indole ring and nitrogen atom.

Phenylalanine: Amino acid with simple benzene ring.

20) Cis-isomerism

Cis-isomerism refers to the arrangement of atoms or groups on the same side of a double bond in a molecule. It is important because it affects the physical and chemical properties of the molecule, such as its melting point, boiling point, and reactivity.

21) Mention the name and symbol of alanine arginine

The name and symbol of alanine are “Alanine” and “Ala” respectively. For arginine, the name and symbol are “Arginine” and “Arg” respectively.

22) Allosteric hemoglobin

Allosteric hemoglobin refers to the ability of hemoglobin to undergo conformational changes in response to the binding of molecules, such as oxygen or carbon dioxide, at sites other than the active site. This allows for the regulation of oxygen binding and release in different tissues.

23) Eicosanoid

Eicosanoids are a large group of lipid messengers with potent effects on every tissue in the body. Eicosanoids are derived from metabolism of 20- carbon, polyunsaturated fatty acids (eicosanoic acids). Eicosanoids are derived from either Omega-3 (ω-3) or Omega-6 (ω-6) fatty acids. Arachidonic acid is the most common precursor of the eicosanoids.

24) Structure of the amino acid

The structure of an amino acid includes a central carbon atom (α-carbon) bonded to an amino group (NH2), a carboxyl group (COOH), a hydrogen atom (H), and a variable side chain known as the R group. The R group determines the specific characteristics of each amino acid.

25) Bond between DNA strand

The bond that holds the two strands of DNA together is a hydrogen bond. These hydrogen bonds form between complementary nitrogenous bases: adenine (A) with thymine (T) and cytosine (C) with guanine (G). This base pairing is a key structural feature of the DNA double helix.

26) What is simple lipids discuss its two types

Simple lipids:

Esters of fatty acids with various alcohols

These contain:

a. Fats (and Oils) and
b. Waxes.

Fats: Esters of fatty acids with glycerol (Oils are fats in the Liquid state)

Waxes: Esters of fatty acids with higher molecular weight monohydric alcohols. (having one OH group)

27) Waxes

Esters of fatty acids with higher molecular weight monohydric alcohols(having one OH group).

28) Types of waxes

• Waxes are of two types:

1) ***True Waxes*** • Waxes are solid, simple lipids containing a monohydric alcohol (with a higher molecular weight than glycerol)
2) ***Other Waxes or Non true waxes*** • include esters of: • Cholesterol • Vitamin A • Vitamin D

29) Bees wax and spermaceti

• Waxes are widely distributed in nature such as the secretion of certain insects as; • Bees-wax, • Spermaceti of the sperm whale

Bees-wax is secreted by the honeybees that use it to form the combs. • It is a mixture of waxes • chief constituent is myricyl (30C), palmitate (16C)

Spermaceti is a wax that is primarily found in the head cavities of sperm whales. It consists of fatty esters, mainly cetyl palmitate and cetyl myristate. Spermaceti has been used in cosmetics, pharmacy, and candles, but its use has been restricted due to international regulations on whale captures. Synthetic cetyl palmitate is now used as a replacement.

30) Five properties of waxes

- Waxes are insoluble in water, but soluble in fat
- Solvents and are negative for acrolein test.
- Very resistant to rancidity.
- Waxes are not easily hydrolyzed as the fats
- waxes are indigestible by lipases (enzymes responsible for fat Digestion in body) Thus they are of no nutritional value

31) what is sterol & write example

Sterols are a type of lipid that have a specific structure, including a four-ring core. They are found in cell membranes and are important for maintaining membrane fluidity and stability. Cholesterol is a well-known example of a sterol, which is found in animal cell membranes and serves various functions in the body, including hormone synthesis and cell signaling.

32) Cholesterol amphiphilic or amphipathic

"Amphipathic" and "amphiphilic" are often used interchangeably to describe molecules that have both hydrophilic and hydrophobic regions in their structure. Cholesterol is considered amphiphilic, as it has both hydrophilic (water-attracting) and hydrophobic (water-repelling) parts in its molecular structure. This characteristic enables it to interact with both water-soluble and lipid-soluble molecules in biological systems.

33) Globosides

They are glycosphingolipids • with two or more sugars, usually

- D -glucose,
- Dgalactose, or
- N-acetyl-D galactosamine.

Globosides are also found in human serum, spleen and liver. A glycosphingolipid that has only one sugar as the side chain is called a ***cerebroside*** • Globosides are highly abundant in RBCs.

34) Iodine number & it's importance?

Iodine number is defined as the number of grams of iodine absorbed by 100 gm of fat.

Importance:

The iodine number is important because it helps determine the degree of unsaturation in fats, which can indicate the quality of an oil and its susceptibility to oxidation and rancidification. It also helps identify potential adulteration in oils.

Or

• The more the iodine number, the greater the degree of unsaturation. • Fats rich in saturated fatty acids have low iodine numbers, • while fats rich in unsaturated fatty acids have high iodine numbers • The determination of iodine number is useful to the chemist in determining the quality of an oil or its freedom from adulteration. • Iodine number of cotton seed oil varies from 103 to 111. • That of olive oil from 79 to 88, • And that of linseed oil from 175 to 202. • A commercial lot of olive oil which has an iodine number higher than 88 might have been adulterated with cotton seed oil • The higher is the iodine number, • the more reactive, • less stable, • more susceptible to oxidation and • rancidification is the oil or fat.

35) Rancidity and factors causing rancidity?

• It is a physico-chemical change in the natural properties of the fat leading to the development of unpleasant odor or taste or abnormal color. It occurs particularly

- on aging
- after exposure to atmospheric oxygen,
- light,
- moisture,
- bacterial or fungal contamination
- and/or heat.

Rancidity is due to

I. Oxidation
II. Hydrolysis

36) Difference between oxidative and hydrolytic rancidity

- **Oxidative Rancidity**

Oxidative rancidity occurs when fats undergo oxidation, resulting in the formation of aldehydes, ketones, and dicarboxylic acids that contribute to an objectionable taste and odor. This type of rancidity requires the presence of oxygen in the air and can be prevented by adding antioxidants like vitamin E to foods.

- **Hydrolytic Rancidity**

Hydrolytic rancidity, on the other hand, is caused by the slow hydrolysis of fats. In the case of fats like butter, this leads to the release of short-chain fatty acids that are volatile and have a rancid taste and odor. Unlike oxidative rancidity, hydrolytic rancidity does not require the presence of oxygen.

37) Role of TAG (triacylglycerols) in plants

Triacylglycerols (TAGs) play an essential role in plants as a storage form of energy. They are stored as oils in the seeds of many plant species. During seed germination, TAGs are broken down by enzymes called lipases, releasing fatty acids that provide energy and serve as building blocks for biosynthesis. This process allows the plant to utilize the stored energy and precursors for growth and development. Additionally, lipases help transport fatty acids to other parts of the plant where they are needed for various metabolic processes.

38) Discuss glycerol

Glycerol:

• It is a simple poly hydroxy alcohol (also called polyol or sugar alcohol) and part of a class of Lipids; glycolipids.

• It contains 3 carbons and 3 hydroxyl (OH) groups. Glycerol is synthesized from dihydroxyacetone phosphate (an intermediate of the glycolytic pathway)

• Dihydroxyacetone phosphate is acted upon by two enzymes to form glycerol, namely Glycerol Phosphate Dehydrogenase and Glycerol Kinase respectively.

• Glycerol is a precursor for synthesis of triacylglycerols and of phospholipids in the liver and adipose tissue. When the body uses stored fat as a source of energy, glycerol and fatty acids are released into the bloodstream and enter the glycolysis pathway directly

39) what is non polar?& R are alkyl group

Nonpolar molecules lack a significant separation of electric charge and do not readily dissolve in polar solvents like water. When the letter "R" is used in chemical structures, it represents a generic alkyl group, which is a group of carbon and hydrogen atoms bonded together in a chain or a branched structure. Alkyl groups are often nonpolar, contributing to the overall nonpolarity of a molecule.

40) How amino acid differ from each other?

Amino acids differ from each other based on their side chains, also known as R-groups. The side chains can vary in size, shape, charge, and chemical properties. These differences in side chains give each amino acid its unique characteristics and determine its role in protein structure and function.

41) Negative amino acid

Negative amino acids are a group of amino acids that have a negatively charged side chain at physiological pH due to the presence of a carboxyl group (COO^-). The two commonly referred to negative amino acids are aspartic acid (Asp) and glutamic acid (Glu). Their side chains are aspartate and glutamate, respectively. These negatively charged side chains can participate in various biochemical interactions, such as forming salt bridges with positively charged amino acids or binding to metal ions in proteins.

42) Amino acids in beta bond

Amino acids in a beta bond refer to the peptide bond formed between two amino acids in a protein chain. This bond is also known as a beta peptide bond. It is formed through a condensation reaction between the carboxyl group (-COOH) of one amino acid and the amino group (-NH2) of another amino acid, resulting in the release of a water molecule. This bond is crucial for the formation of the polypeptide chain in proteins.

43) What is Mediterranean diet?

The Mediterranean diet is a type of diet that is commonly followed in Mediterranean cultures. It is known for its association with a lower risk of heart disease. This diet is rich in monounsaturated fatty acids from sources like olive oil and omega-3 fatty acids from fish oils and nuts. It is low in saturated fat and typically includes fresh seasonal foods, a variety of plant-based foods, limited red meat consumption, and the use of olive oil as the main source of fat. Compared to a typical Western diet high in saturated fats, the Mediterranean diet has been found to lower total cholesterol and LDL cholesterol levels, with little effect on HDL cholesterol levels.

44) Pentose Sugar of Nucleic Acid?

• D-ribose and 2-deoxy D-ribose are the only sugars so far found in the nucleic acids. • These aldo pentoses belong to D-family • They are present as Furanose (ring) in the form of β-Anomer. • The addition of a pentose sugar to a base produces a nucleoside

- If the sugar is Dribose, a ribonucleoside is produced
- If the sugar is 2 deoxy Dribose, a deoxyribonucleoside is produced

45) 3 Function of cGMP

Cyclic GMP is synthesized from GTP • It serves as a second messenger in response to nitric oxide during relaxation of smooth muscle (especially blood vessels) so it has role in smooth muscle relaxation and vasodilatation. It also has role in

- Protein phosphorylation
- Neurotransmission
- Insulin action
- Regulation of sodium channels

46) Nucleotide composition

Nucleotides are composed of a nitrogenous base (purine or pyrimidine), A pentose monosaccharide, and One, two, or three phosphate groups.

The nitrogenous bases belong to two families of compounds: Purines and Pyrimidines. Both DNA and RNA contain the same purine bases: Adenine (A) and Guanine (G).

Pyrimidines include:

1) Cytosine C—in both DNA and RNA

2) Thymine (T)—only in DNA
3) Uracil (U) —only in RNA

47) How carbon atoms are numbered in nucleotides and nucleosides?

In nucleotides and nucleosides, the carbon atoms in the pentose sugar are numbered 1' to 5', with the prime symbol ('), to distinguish them from the numbered atoms of the nitrogenous base. The prime symbol is used to differentiate the sugar atoms from the atoms in the heterocycle (nitrogenous base). So, when referring to the 5'-carbon of a nucleoside or nucleotide, it is specifying a carbon atom in the pentose sugar, not in the base.

48) Define the medical properties of nucleotides and nucleic acids?

Specifically medical applications include the use of synthetic purine and pyrimidine analogs that contain halogens, thiols, or additional nitrogen atoms.

There use include chemotherapy for cancer as suppressors of the immune response during organ transplantation.

As anti-viral drugs such as in the treatment of AIDS.

49) Cofactor

A cofactor is a molecule or ion that helps enzymes function properly. • Some enzymes require no chemical groups for activity other than their amino acid residues. • Whereas some enzymes require molecules other than proteins for enzymic activity. • If the non-protein moiety is a metal ion such as Zn2+ or Fe2+, it is called a cofactor. ***For example,*** the metal ion magnesium serves as a cofactor for the enzyme ATP synthase, which is involved in energy production in cells.

50) Intermediate enzyme

An intermediate enzyme typically refers to an enzyme that is part of a metabolic pathway and acts on a specific substrate to convert it into an intermediate product. These intermediate products are then further acted upon by other enzymes in the pathway until the final desired product is produced. Intermediate enzymes play a crucial role in regulating metabolic pathways and ensuring that the process occurs in a stepwise manner.

51) Examples of hydrolases/ Name some hydrolase enzymes

Hydrolases:

These enzymes catalyze hydrolysis, i.e. • add water molecule to the substrate which is simultaneously decomposed; the functional group of substrate is transferred to water.

Common ***example*** of hydrolases are: • Protein hydrolyzing Enzymes (peptidases). • Carbohydrases • Lipid hydrolyzing enzymes e.g. Lipases and • Phospholipases.

52) Buffer and it's composition?

A buffer is a solution that resists changes in its pH when an acid or base is added to it. It consists of a weak acid and its conjugate base, or a weak base and its conjugate acid. This combination helps maintain the solution's pH by absorbing or releasing hydrogen ions as needed, which prevents drastic shifts in acidity or alkalinity.

53) Relationship of velocity to enzyme concentration

• The rate of the reaction is directly proportional to the enzyme concentration. • There is a linear relationship between reaction rate and enzyme concentration (at constant substrate concentration). ***For example,*** • if the enzyme concentration is halved, both the initial rate of the reaction (Vo) and the maximum rate (Vmax) will be reduced to half of their original values.

54) What is reaction velocity & how it affects th rate of reaction?/Reaction Rates and Order of Reactions/Reaction Velocity (v)

• The rate or velocity of a reaction (v) is the number of substrate molecules converted to product per unit time; • Velocity is usually expressed as μmol of product formed per minute. The rate of enzyme catalyzed reaction increases as the substrate concentration increases until it reaches a maximum velocity (Vmax). This maximum velocity means that all the available binding sites on the enzyme are occupied by substrate molecules, and further increases in substrate concentration do not affect the reaction rate.

55) Order of reaction

When [S] is much ***less***, then the velocity of the reaction is approximately proportional to the substrate concentration. • The rate of reaction is then said to be first order with respect to substrate.

• When [S] is much ***greater*** than Km the velocity is constant and equal to Vmax. • The rate of reaction is then independent of substrate concentration, and is said to be zero order with respect to substrate concentration

56) Zero Order Reaction

In chemistry, a zero-order reaction is a type of reaction where the rate of the reaction is constant and independent of the concentration of the reactant(s). In other words, the reaction proceeds at a fixed rate, regardless of how much reactant is present. The rate equation for a zero-order reaction can be represented as "Rate = k", where "k" is the rate constant. This means that the concentration of the reactant(s) does not affect the rate of the reaction.

57) Induce fit hypothesis

Some proteins can change their shape (conformation). When a substrate Combines with an enzyme, it induces a change in the enzyme's conformation. This change in Conformation when the substrate binds is Induced by multiple weak interactions with the substrate. There may also be rearrangements of covalent bonds during an enzyme-catalyzed reaction. This conformational change is referred to as induced fit.

58) Km low or high

- **Small Km:**

A numerically small (low) Km reflects a high affinity of the enzyme for substrate, because a low concentration of substrate is needed to half-saturate the enzyme—that is, to reach a velocity that is 1⁄2 Vmax.

- **Large Km:**

A numerically large (high) Km reflects a low affinity of enzyme for substrate because a high concentration of substrate is needed to half-saturate the enzyme.

59) Write equation of free energy

The equation for free energy (Gibbs free energy) is:

G = H - TS

where:

(G) represents the Gibbs free energy

(H) is the enthalpy (total heat content) of the system

(T) is the absolute temperature

(S) is the entropy (measure of disorder) of the system.

The equation for free energy is $\Delta G = \Delta H - T\Delta S$,

where ΔG is the change in free energy, ΔH is the change in enthalpy, T is the temperature in Kelvin, and ΔS is the change in entropy.

60) What do you know about km of michaelis- menton equation?

Michaelis-Menten equation, the rate equation (Contd.)

$V_0 = V_{max}[S] / Km + [S]$

• Where; • V_0 = initial reaction velocity • V_{max} = maximal velocity • Km = Michaelis constant • = K1+K2 / K-1 • [S] = substrate concentration

• This is the Michaelis-Menten equation statement of the quantitative relationship between the; • initial velocity V_0, • the maximum velocity V_{max}, and • the initial substrate concentration [S], all related through the Michaelis constant Km.

61) What is the concentration of OH in a solution with an H+ concentration of 1.3x 10-4M?

Solution:

- $Kw = [H^+][OH^-]$
- With [H+] = 1.3x 10-4M, solving for [OH-] gives
- [OH-] = Kw/[H+]
- $= 1x10^{-14}M^2 / 1.3x\ 10^{-4}M$
- $= 7.7 \times 10^{-11}$

62) What is the concentration of H+ in a solution of 0.1 M NaOH?

Solution:

Kw= $[H^+][OH^-]$

• With [OH-] = 0.1 M, solving for [H+] gives

• $[H^+] = Kw/[OH^-] = 1 \times 10^{-14}M^2/ 0.1M$

$= 10^{-14}M^2/0.1M$

$= 10^{-13}M$

63) Calculate the ratio of the concentrations of Acetate and acetic acid required in a buffer System of pH 5.30 (pKa = 4.76)

pH= pKa + log [acetate]/[acetic acid]

log [acetate]/[acetic acid]= pH- pKa

= 5.30-4.76

= 0.54

[acetate]/[acetic acid] = antilog 0.54

= 3.5

64) Biomedical importance of nucleotides(any 3)

1) ***Precursors of nucleic acids:*** Nucleotides are the building blocks of nucleic acids. • Without them, DNA or RNA can not be produced.
2) ***Transmission of genetic information;*** • This gives them the ability to store and transmit genetic information from; • one generation to the next which is a fundamental condition for life
3) ***Protein synthesis:*** Protein synthesis relies on nucleotides as they hold the key to expressing genetic information.
4) ***Energy currency:*** Nucleotides play an important role as "energy currency" in the cell. • Nucleoside tri and diphosphates such as ATP and ADP are the principal donors and acceptors of phosphoryl group in metabolism. • By doing this, they play a key role in the energy transduction. • This energy is used in almost every energy requiring process in the body, such as; • Muscle contraction, • Transmission of nerve impulse, • Transports of nutrients across cell membrane • Motility of spermatozoa • And many more energy dependent processes .
5) ***Signal Transduction:*** GTP and GDP play key • roles in activating or inhibiting proteins in • various cellular signaling cascades.
6) ***Medical applications***: Specifically medical applications include the
    a. use of synthetic purine and pyrimidine analogs
    b. that contain halogens, thiols, or additional nitrogen atoms;
    c. There use include chemotherapy for cancer
    d. as suppressors of the immune response during organ transplantation.
    e. as anti-viral drugs such as in the treatment of AIDS.

65) Characteristics of glycerol

- It is a simple poly hydroxy alcohol (also called polyol or sugar alcohol) and part of a class of Lipids; glycolipids.
- It contains 3 carbons and 3 hydroxyl (OH) groups. Glycerol is synthesized from dihydroxyacetone phosphate (an intermediate of the glycolytic pathway)
- Dihydroxyacetone phosphate is acted upon by two enzymes to form glycerol, namely Glycerol Phosphate Dehydrogenase and Glycerol Kinase respectively.
- Glycerol is a precursor for synthesis of triacylglycerols and of phospholipids in the liver and adipose tissue. When the body uses stored fat as a source of energy, glycerol and fatty acids are released into the bloodstream and enter the glycolysis pathway directly

66) Enlist hydrolysis of maltose, lactose and sucrose

Maltose: Hydrolysis of maltose results in the formation of two glucose molecules.

Maltose + H2O → 2 glucose

Lactose: Hydrolysis of lactose yields one molecule of glucose and one molecule of galactose.

Lactose + H2O → glucose + galactose

Sucrose: Hydrolysis of sucrose produces one molecule of glucose and one molecule of fructose.

Sucrose + H2O → glucose + fructose

67) Define Immunoglobulin, conjugated proteins, and complex protein?

Immunoglobulins: Immunoglobulin, also known as antibodies, are proteins produced by the immune system that serve as the first line of defense against bacteria and viruses and other foreign agents.

Conjugated proteins are proteins that are combined with non-protein molecules, such as lipids or carbohydrates.

Complex proteins are proteins made up of multiple polypeptide chains or subunits. Examples include hemoglobin and collagen.

LONG QUESTIONS(5)

68) pKa Value

• pKa is the measure of the tendency of a group to give up (donate) a proton with the tendency decreasing ten-fold as the pKa increases by one unit • Each titratable group has a pKa that is numerically equal to the pH at which exactly one half of the protons have been removed from that group • The pKa for the most acidic group (COOH) is pK1, whereas the pKa for the next most acidic group (NH3+) is pK2 • In acid solution (e.g., pH 1), the amino group is protonated (–NH3+) and the carboxyl group is not dissociated (–COOH) • As the pH is raised, the carboxylic acid is the first group to give up a proton, as its pKa is near 2 • The dipolar form persists until the pH approaches 9, when the protonated amino group loses a proton.

69) Action of acids on carbohydrates

• Polysaccharides are hydrolyzed into their constituent monosaccharides by boiling with dilute acids • With concentrated acids the monosaccharides are decomposed • Pentoses yield a cyclic aldehyde furfural with 12% HCL • Hexoses are decomposed by hot strong acids to give hydroxy methyl furfural. • The furfural products thus formed can condense with certain organic phenols (αnaphthol) to form compounds having characteristic colours • This forms basis of certain tests used for detection of sugars • Molisch,s and • Selivanoff,s are examples of such tests.

70) N-glycosidic bond & How it forms?

An N-glycosidic bond is a covalent bond, type of glycosidic bond, that connects a nitrogenous base to a sugar molecule, forming a nucleoside. The bond forms between the nitrogen atom (N) of the nitrogenous base and the anomeric carbon atom of the sugar. This bond is formed through a condensation reaction where a water molecule is eliminated.

In DNA and RNA, for instance, N-glycosidic bonds link the nitrogenous bases (adenine, guanine, cytosine, thymine, or uracil) to the sugar backbone. The specific type of N-glycosidic bond depends on the base and sugar involved. For example, in DNA, adenine forms an N-glycosidic bond with deoxyribose sugar. N-glycosidic bonds are important for the structure and function of biomolecules, including DNA, RNA, and various glycoproteins found in cells.

71) Beta-bends/ Beta conformation of protein

Beta-bends are short regions that consist of four consecutive amino acid residues, and they frequently connect strands of antiparallel betasheets. They help the polypeptide chain reverse its direction, which is essential for it to form a compact, globular shape. The structure of a beta-bend involves a 180-degree turn that is made up of four amino acid residues. The carbonyl oxygen of the first residue forms a hydrogen bond with the amino-group hydrogen of the fourth residue. The name "beta-bends" was given because they are often located between successive strands of antiparallel β sheets.

72) Primary structure of protein

The primary structure of a protein refers to the linear sequence of amino acids in the polypeptide chain. • The primary structure is held together by covalent bonds such as peptide bonds, which are made during the process of protein biosynthesis or translation. The two ends of the polypeptide chain are referred to as the carboxyl terminus (C-terminus) and the amino terminus (N-terminus) based on the nature of the free group on each extremity. The primary structure of a protein determines how it folds up into its unique three-dimensional structure, and this in turn determines the function of the protein. Understanding of the primary structure of proteins is important because many genetic diseases result in proteins with abnormal amino acid sequences which cause loss or impairment of normal function.

73) Secondary structure of Protein

• Secondary structure refers to particularly stable arrangements of amino acid residues giving rise to recurring structural patterns

The primary structure of a protein folds into secondary structures such as alpha helices and beta sheets, which are stabilized by hydrogen bonds between the backbone atoms.

• Secondary structure, the folding of short (3to 30-residue), contiguous segments of polypeptide into geometrically ordered units • The term secondary structure refers to any chosen segment of a polypeptide chain and describes the local spatial arrangement of its main-chain atoms, without regard to its relationship to other segments • There are a few types of secondary structure that are

particularly stable and occur widely in proteins • The most prominent are the α helix and β conformations.

Alpha helices are formed by a right-handed coiled structure, while beta sheets are formed by multiple strands of polypeptide chains that are aligned in a parallel or antiparallel manner. The secondary structure of a protein is important because it helps determine the overall three-dimensional structure and function of the protein.

74) Forces involved in three dimensional structure of proteins

Molecular forces that stabilise three dimensional tertiary structure include:

1) Electrostatic Interactions
2) Hydrophobic interactions
3) Disulphide bridges
4) Hydrogen bonding

Electrostatic Interactions: • The positive charges on the basic amino acids enables them to form ionic bonds (electrostatic bonds) with negatively charged groups, such as the side chains of acidic amino acids.

Hydrophobic Interactions: • Hydrophobic Interactions occur between non-polar uncharged side chains such as between aromatic groups of phenylalanine side chains.

Disulfide Bridges: • The amino acid cysteine in a protein can form a covalent disulfide bond with another cysteine molecule through spontaneous (nonenzymatic) oxidation of their sulfhydryl groups.

Hydrogen bonds: • Hydrogen bonds in which a hydrogen atom is shared by a nitrogen (or oxygen) in the peptide backbone and an oxygen atom in an amino acid side chain.

75) Process of Hydrogenations/Hydrogenation with respect to fats

Hydrogenation reaction involves conversion of a carbon-carbon double bond to a carbon-carbon single bond through the addition of hydrogen.

As you continue to hydrogenate your molecule

- Melting point increases
- Fat becomes more solid at room temp

Oils which are liquid at ordinary room temperature, on hydrogenation become solidified. • This is the basis of Banaspti ghee manufacturing. • Where inedible and cheap oils like cotton seed oil are hydrogenated and converted to edible solid fats. • The hydrogenation is done • under high pressure of hydrogen and is catalyzed by finely divided nickel or copper and heat.

76) Aromatic Amino Acids

• The aromatic amino acids have been grouped together because they all contain ring structures with similar properties, but their polarity differs • The aromatic ring is a six-membered carbon–hydrogen ring with three conjugated double bonds (the benzene ring or phenyl group). Benzene is an organic chemical compound with the chemical formula C6H6. Its molecule is composed of 6 carbon atoms joined in a ring, with 1 hydrogen atom attached to each carbon atom. • The substituents on this ring determine whether the amino acid side chain engages in polar or hydrophobic interactions

• This group includes – Phenylalanine – Tyrosine – Tryptophan – with their aromatic side chains, are relatively nonpolar (hydrophobic). – All can participate in hydrophobic interactions.

77) Chemistry of monoenoic lipid

Monoenoic lipids contain a single double bond in their fatty acid chains. This double bond introduces a kink in the chain, resulting in a lower melting point compared to saturated lipids. Monoenoic lipids are typically liquid at room temperature and are often referred to as "oils." Chemically, the double bond in monoenoic lipids can undergo reactions such as hydrogenation, oxidation, and isomerization. These reactions can affect the stability and reactivity of monoenoic lipids, impacting their shelf life and nutritional properties. The presence of the double bond is a key characteristic of monoenoic lipids, influencing both their physical and chemical properties.

78) Platelet-activating factor (PAF)

PAF (platelet-activating factor) is an unusual type of ether glycerophospholipid. It has a saturated alkyl group connected to carbon 1 through an ether link, and an acetyl residue (instead of a fatty acid) at carbon 2 of the glycerol backbone. PAF is produced and released by various cell types. When it binds to surface receptors, it triggers powerful blood clotting and acute inflammatory responses. PAF induces platelets to clump together and release their contents, which is essential for clot formation. It also stimulates neutrophils and alveolar macrophages to produce superoxide radicals, which are necessary for killing microbes. In simpler terms, PAF is a unique lipid that can cause blood clotting and inflammation by activating different cells in the body.

79) Sugar in DNA and their characteristics

The sugar in DNA is deoxyribose, a five-carbon sugar. It is characterized by the absence of an oxygen atom at the 2' carbon position, which gives it the name "deoxy." This structural difference from ribose, the sugar in RNA, contributes to the stability of the DNA molecule. Deoxyribose forms the backbone of the DNA strand, with the nitrogenous bases (adenine, guanine, cytosine, and thymine) attached to it. The arrangement of deoxyribose and the nitrogenous bases forms the double helix structure of DNA, allowing for the storage and transmission of genetic information.

80) Nucleotide as energy currency or cofactor

As Energy currency: Nucleotides play an important role as "energy currency" in the cell. • Nucleoside triand diphosphates such as ATP and ADP are the principal donors and acceptors of phosphoryl group in metabolism. • By doing this, they play a key role in the energy transduction. • This energy is used in almost every energy requiring process in the body, such as; • Muscle contraction, • Transmission of nerve impulse, • Transports of nutrients across cell membrane • Motility of spermatozoa • And many more energy dependent processes .

As cofactor: Nucleotides can also act as cofactors, which are molecules that assist enzymes in catalyzing biochemical reactions. For example, coenzyme A (CoA), which is derived from the nucleotide adenosine triphosphate (ATP), serves as a cofactor in several metabolic pathways. So, nucleotides function both as energy currency and cofactors in cellular processes.

81) Why the Nucleotide are called regulatory molecules

Nucleotides are important regulatory compounds for many of the pathways of intermediary metabolism, inhibiting Or activating key enzymes.

- Roles that nucleotides perform in metabolic regulation include:
- ATP-dependent enzyme Phosphorylation in key metabolic reactions.
- Allosteric regulation of enzymes by ATP, AMP, and CTP
- Phosphorylation can either activate or inhibit ATP- dependent enzyme depending on the type of enzyme.
- Similar is the case with Dephosphorylation.

82) Effect of pH

• The pH optimum varies for different enzymes: • The pH at which maximal enzyme activity is achieved is different for different enzymes, and often reflects the [H+] at which the enzyme functions in the body. • ***For example,*** pepsin, a digestive enzyme in the stomach, is maximally active at pH 2. • Whereas other enzymes, designed to work at neutral pH, are denatured by such an acidic environment • Another examples is that there are two types of phosphatases in the body. • The one that acts in the alkaline pH is called alkaline phosphatase and • the other which acts at acidic pH is known as acid phosphatase.

83) Effect of pH on the ionization of the active site

• The concentration of H+ affects reaction velocity in several ways. • First, the catalytic process usually requires that the enzyme and substrate have specific chemical groups in either an ionized or un-ionized state in order to interact. • ***For example,*** catalytic activity may require that an amino group of the enzyme be in the protonated form (–NH3 +). • At alkaline pH, this group is deprotonated, and the rate of the reaction, therefore, declines. • Extremes of pH can also lead to denaturation of the enzyme.

84) How activation energy related to enzyme

Enzymes lower the activation energy required for a chemical reaction to occur. Activation energy is the energy needed to start a reaction and break the existing bonds in the reactant molecules. Enzymes facilitate this process by binding to the reactant molecules, creating an enzyme-substrate complex. This complex lowers the activation energy by stabilizing the transition state, allowing the reaction to proceed more quickly. Enzymes achieve this through various mechanisms, such as bringing reactants closer together, providing a favorable microenvironment, or directly participating in the reaction. By reducing the activation energy, enzymes increase the rate of chemical reactions, making them essential for biological processes.

85) What are inorganic elements that are involved in the co-enzymes?

Coenzymes are organic molecules that assist enzymes in catalyzing biochemical reactions. However, some coenzymes contain essential metal ions (inorganic elements) as cofactors. For example:

1. **Zinc (Zn2+):** It is a cofactor in various enzymes, including those involved in DNA replication, RNA transcription, and protein digestion.
2. **Iron (Fe2+ and Fe3+):** Iron ions are found in heme groups of hemoglobin and myoglobin, which are involved in oxygen transport and storage.
3. **Magnesium (Mg2+):** It's required for the activity of numerous enzymes, particularly those involved in ATP metabolism.
4. **Copper (Cu+ and Cu2+):** Copper ions are found in enzymes involved in electron transport, such as cytochrome c oxidase.

5. **Manganese (Mn2+):**It's involved in various enzymes related to metabolism, including those participating in the breakdown of carbohydrates and amino acids.

These metal ions serve as integral parts of coenzymes or cofactors, aiding in the catalytic activity of enzymes by providing structural stability and participating in electron transfer reactions.

86) what information you get from titration of Glycine curve.

The titration curve of glycine provides insights into its acid-base properties. It displays two equivalence points corresponding to the deprotonation of the carboxyl and amino groups at pH values around 2.3 and 9.6, respectively. The buffering regions around these pKa values indicate glycine's ability to resist pH changes when small amounts of acid or base are added. Inflection points on the curve mark where the buffering capacity changes rapidly. The curve helps determine the pH range over which glycine can effectively act as a buffer and reveals its ionization behavior during titration.

87) Isozyme of hexokinase illustrate the significance of Km

• *A comparison between the isozymes of hexokinase illustrates the significance of the Km.* •

Hexokinase I, found in red blood cells, has a low Km for glucose (approximately 0.05 mM). This allows it to efficiently utilize glucose even when blood glucose levels are very low. On the other hand, the isozyme of hexokinase known as glucokinase, found in the liver, has a higher Km (approximately 5 to 6 mM). This higher Km is beneficial for the liver to store excess glucose as glycogen or convert it into fat after a carbohydrate-rich meal. In summary, the different forms of hexokinase have varying affinities for glucose, enabling cells to adapt to different glucose concentrations and perform specific metabolic functions.

88) Functions of oxidoreductase

Oxidoreductases are a class of enzymes that facilitate oxidation and reduction reactions, playing a vital role in cellular metabolism. Their primary function is to transfer electrons between molecules, resulting in the conversion of one molecule into another while conserving energy. Oxidoreductases are essential in processes like energy production (as in the electron transport chain), detoxification of harmful substances, and biosynthesis of molecules such as lipids and amino acids.

89) Palmitic Acid(16:0) Characteristics

- Palmitic acid, or hexadecanoic acid, is the most common saturated Fatty Acid found in animals, plants and microorganisms
- Palmitic acid mainly occurs as its ester in triglycerides (fats), especially in palm oil.
- It is also found in high amounts in • Butter, • Cheese, • milk and • meat
- Excess carbohydrates in the body are converted to palmitic acid.
- Palmitic acid is the first fatty acid produced during fatty acid synthesis and the precursor to longer fatty acids.

90) Features of VLDL

- VLDL is a type of lipoprotein that transports triglycerides from the liver to peripheral tissues.
- It is composed of triglycerides, cholesterol, and proteins.
- VLDL is a precursor to LDL (Low-Density Lipoprotein), also known as "bad cholesterol."
- High levels of VLDL in the blood are associated with an increased risk of cardiovascular disease.
- VLDL levels can be influenced by factors such as diet, genetics, and certain medical conditions.
- VLDL is synthesized in the liver and released into the bloodstream.
- It provides energy to cells by delivering triglycerides to tissues for storage or immediate use.
- VLDL particles are larger and less dense than LDL particles.
- VLDL levels can be measured through a blood test called a lipid profile.
- Managing VLDL levels through lifestyle changes, such as a healthy diet and regular exercise, can help maintain cardiovascular health.

91) Function of cAMP

Definition:

Cyclic AMP is a cyclic nucleotide • cAMP is synthesized in tissues from ATP • chemically it is 3´-5´ adenosine monopohosphate.

Function:

• Acts as second messenger in the cell • It has role in glycogen metabolism • ↑cAMP, ↑glycogenolysis • ↑cAMP ↑TAG metabolism • ↑cAMP ↑ lipolysis • It decreases cholesterol synthesis • It causes activation of protein kinases which in turn; • activate or deactivate other enzymes. • It regulates the cell membrane permeability, by increasing permeability of cell membrane to H2O, Na+, K+& Ca+2 • Moreover, it regulates • insulin secretion, • catecholamine biosynthesis & • Melatonin synthesis.

92) Properties of TAGs

- **Physical properties**

• Neutral fats are

1) colourless,
2) odorless and
3) tasteless substances
4) ***Solubility***: • They are insoluble in water but • soluble in organic fat solvents(e.g., ether, benzene, acetone, chloroform) •
5) ***Specific gravity:*** • The specific gravity of all fats is less than 1.0, • consequently all fats float in water . Specific gravity of water is 1 •
6) ***Emulsification:*** • Emulsions of fat may be made by shaking vigorously in water and by emulsifying agents such as gums and soaps • The emulsification of dietary fats in intestinal canal, brought about by bile salts, is • a prerequisite for digestion and absorption of fats. • The bile salts, act to break apart the fat globules in the small intestines and allow them to become more "soluble" for absorption. • The hydrophobic fat molecules will clump together into globules in the watery mixture in the digestive system. • The emulsifiers break them down to smaller "globules" and allow them to become more soluble.

93) Five products of protein

Proteins serve as crucial components in various biological processes. Five important products of proteins include:

- ***Enzymes***: Proteins catalyze biochemical reactions as enzymes, accelerating chemical processes within cells.
- ***Structural Proteins***: Proteins like collagen and keratin provide structural support to tissues, including skin, hair, and connective tissues.
- ***Hormones***: Certain proteins act as hormones, regulating physiological processes such as insulin controlling blood sugar levels.
- ***Antibodies***: Proteins in the immune system, known as antibodies, target and neutralize foreign invaders like bacteria and viruses.
- ***Transport Proteins:*** Proteins like hemoglobin transport oxygen in blood, while others facilitate the movement of molecules across cell membranes.

94) Proteins classify on which basis?

Proteins can be classified based on several different criteria, including:

- **Structure**: Proteins can be classified by their structural complexity, such as primary, secondary, tertiary, and quaternary structures.
- **Function:** Proteins can be grouped by their roles and functions, such as enzymes, structural proteins, transport proteins, antibodies, and hormones.
- **Shape**: Proteins can be categorized based on their three-dimensional shapes, such as globular (spherical) or fibrous (elongated) proteins.
- **Composition**: Proteins can be classified by their amino acid composition, distinguishing them based on the types and sequence of amino acids they contain.
- **Solubility**: Proteins can be categorized as water-soluble (globular) or water-insoluble (fibrous) based on their solubility characteristics.
- **Source**: Proteins can be classified based on their source of origin, such as animal, plant, or microbial proteins.
- **Post-Translational Modifications:** Proteins can also be grouped by the presence of specific post-translational modifications, which can affect their structure and function.

95) Carbohydrates and their types

• Carbohydrates are polyhydroxy aldehydes or ketones, or substances that yield such compounds on hydrolysis • Many, but not all, carbohydrates have the empirical formula$(CH_2O)_n$; some also contain nitrogen, phosphorus or sulfur.

Classification

Carbohydrates are classified into four major groups:

- Monosaccharides
- Oligosaccharides
- Disaccharides
- Polysaccharides

• **Monosaccharides** (simple sugars) are those which cannot be hydrolyzed further into simpler forms.p

- Glucose

• **Disaccharides** are those sugars which yield two molecules of the same or different monosaccharides on hydrolysis

– Maltose - 2 molecules of glucose on hydrolysis

• **Oligosaccharides** are short chains of monosaccharide units or residues (3-10) joined by characteristic linkages called glycosidic linkage.

– Raffinose (a trisaccharide composed of glucose, fructose, and galactose)

• **Polysaccharides** (Glycans) are those which yield more than 10 molecules of monosaccharides on hydrolysis • Some have hundreds or thousands of units •

– Some polysaccharides, such as cellulose, are linear chains; others, such as glycogen, are branched

96) Importance of glucose in human body

- 1) Glucose is the primary source of energy for the human body, providing fuel for cellular processes.
- 2) It plays a crucial role in brain function, as the brain relies heavily on glucose for energy.
- 3) Glucose is necessary for the production of ATP, the energy currency of cells.
- 4) It is involved in the regulation of blood sugar levels, with insulin helping to control glucose levels in the bloodstream.
- 5) Glucose is also a precursor for the synthesis of other important molecules, such as glycogen, amino acids, and fatty acids.

97) Lactose and it's properties

Lactose is a disaccharide sugar composed of two monosaccharides, glucose and galactose. It is commonly found in milk and dairy products. Lactose is notable for its β-glycosidic bond between glucose and galactose, which can be broken down by the enzyme lactase. Some key properties of lactose include its role as an energy source, its limited sweetness compared to other sugars, and its potential to cause digestive issues in individuals with lactose intolerance. Lactose is widely used in the food industry as a flavor enhancer, bulking agent, and for its interactions with proteins and other compounds.

98) Cyclooxygenase pathway

Cyclooxygenase Pathway; contain Prostaglandins and Thromboxanes.

Prostaglandins: Prostaglandins are fatty acids containing 20 carbon atoms, including an internal 5-carbon ring. Prostaglandins have a hydroxyl group at C 15, a double bond between C 13 and C 14, and various substituents on 5 membered ring at C9 and C11. Double bonds also may be present between carbons 5 and 6 and between carbons 17 and 18 in case of series 2 and 3.

Thromboxanes (TX): also formed via the cyclooxygenase Pathway, differ from PGs in that they contain a 6- membered ring that includes an oxygen atom. Thromboxanes have a six membered ring containing an oxygen atom. Substituents are attached to the ring at carbons 9 and 11. In the case of TXA2(shown above), an oxygen atom connects carbons 9 and 11.

99) Name pentose sugars in nucleic acid and their characteristics?(2+3)

Pentose Sugar

- D-ribose and 2-deoxy d-ribose are the only sugars so far found in the nucleic acids. These aldo pentoses belong to D-family
- They are present as Furanose (ring) in the form of β-Anomer. The addition of a pentose sugar to a base produces a nucleoside
- If the sugar is Dribose, a ribonucleoside isProduced

Characteristics:

- Pentose sugars have a five-carbon structure, forming the backbone of nucleic acids.
- They exist in a ring form known as furanose, with the fifth carbon bonding to an oxygen atom.
- Hydroxyl (OH) groups are present on various carbon atoms, providing bonding sites.
- DNA contains 2-deoxy-D-ribose, lacking an oxygen at the 2' carbon for stability.
- Pentose sugars link nucleotide units through phosphodiester bonds, shaping DNA and RNA.

100) Nomenclature of prostaglandins

Prostaglandins: Prostaglandins are fatty acids containing 20 carbon atoms, including an internal 5-carbon ring. Prostaglandins have a hydroxyl group at C 15, a double bond between C 13 and C 14, and various substituents on 5 membered ring at C9 and C11. Double bonds also may be present between carbons 5 and 6 and between carbons 17 and 18 in case of series 2 and 3.

Nomenclature of Prostaglandins (PGs) involves the assignment of a capital letter (PGE), a numeral subscript (PGE1), and for the PGF family, a Greek letter subscript (PGF2α). The capital letter refers to the ring substituents at positions X and Y. The numeral subscript that follows the capital letter(e.g. PGE1) refers to the PG series 1, 2, or 3, determined by the number of unsaturated

bonds present in the linear portion of the hydrocarbon chain. It does not include double bonds in the internal ring. The double bonds between carbons 13 and 14 are trans; the others are cis. The Greek letter subscript, found only in the F series, refers to the position of the hydroxyl group at carbon 9. This hydroxyl group primarily exists in the alpha position, i.e. it lies below the plane of the ring.

101) Explain types of RNA

There are several types of RNA that play different roles in cellular processes:

1. ***Messenger RNA (mRNA):*** mRNA carries the genetic information from DNA to the ribosomes, where it serves as a template for protein synthesis.

2. ***Transfer RNA (tRNA):*** tRNA molecules bring amino acids to the ribosomes during protein synthesis. Each tRNA molecule is specific for a particular amino acid.

3. ***Ribosomal RNA (rRNA):*** rRNA is a major component of ribosomes, which are the cellular machinery responsible for protein synthesis. It helps in the assembly of amino acids into polypeptide chains.

4. ***Small nuclear RNA (snRNA):*** snRNA is involved in RNA splicing, a process that removes non-coding regions (introns) from pre-mRNA to produce mature mRNA.

5. ***Small nucleolar RNA (snoRNA):*** snoRNA guides chemical modifications of rRNA and snRNA molecules, ensuring their proper functioning.

6. ***MicroRNA (miRNA):*** miRNA regulates gene expression by binding to mRNA molecules and inhibiting their translation or promoting their degradation.

7. ***Long non-coding RNA (lncRNA):*** lncRNA molecules have diverse functions, including regulation of gene expression, chromatin remodeling, and modulation of protein activity.

102) Nucleic acid characteristics

- Nucleic acids, such as DNA and RNA, are composed of building blocks called nucleotides.

- DNA is a double-stranded helical structure, while RNA is typically single-stranded.

- Nucleic acids store genetic information and are essential for inheritance and evolution.

- Nucleic acids are involved in protein synthesis through the process of transcription and translation.

- DNA replication ensures the accurate copying of genetic information during cell division.

- Nucleic acids can undergo mutations, which can lead to genetic variations and diseases.

- Nucleic acids have unique base pairing rules: adenine (A) pairs with thymine (T) in DNA, and adenine (A) pairs with uracil (U) in RNA, while cytosine (C) pairs with guanine (G) in both DNA and RNA.

- Nucleic acids have negative charges due to the phosphate groups in their structure.

- RNA has various types, including messenger RNA (mRNA), transfer RNA (tRNA), and ribosomal RNA (rRNA), each with specific functions in protein synthesis.

103) DNA interaction

The purine and Pyrimidine bases are Hydrophobic and Relatively insoluble in Water at the near neutral pH of the cell.

• Hydrophobic stacking Interactions in which Bases are positioned With the planes of their rings parallel. Hydrophobic Stacking are an Important interaction Between bases in Nucleic acids.

• The stacking also Involves a Combination of van Der Waals and Dipole-dipole Interactions between the bases. Base stacking helps to minimize contact of the bases with water.

• Therefore base stacking interactions are very important in stabilizing the three dimensional structure of nucleic acids.

• The most important Functional groups of Pyrimidines and Purines are ring nitrogens, carbonyl groups, and exocyclic amino groups.

• Hydrogen bonds involving the amino and carbonyl groups are the second Important mode of Interaction between Bases in nucleic acid molecules.

• The most important Hydrogen-bonding

Patterns are those Defined by James D.

Watson and Francis Crick, in which

• A bonds specifically to T (or U) and G bonds to C. These two types of Base pairs Predominate in double-stranded DNA and RNA.

104) Properties of nitrogenous Bases

• **Aromatic**: The nitrogen containing bases are aromatic i.e. they have alternate double bonds

• **Heterocyclic:**

They are heterocyclic i.e. structures that contain other atoms in addition to carbon, such as nitrogen in the ring structure. The six-atom rings of Purines and pyrimidines are numbered in opposite directions.

• **Weak Bases:** Purines or pyrimidines with an –NH2 group are weak bases.

Functional Groups: The most important functional groups of pyrimidines and purines are ring nitrogens, carbonyl groups

And exocyclic amino groups

• **Hydrophobicity**:

The purine and Pyrimidine bases are hydrophobic and relatively insoluble in water at the near neutral cell pH.

• **Stacking Interaction:** Hydrophobic stacking interactions in which two or more bases are positioned with the planes of their rings parallel (like a stack of coins) are one of two important modes of interaction between bases in nucleic acids.

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